🏀 Basketball Shot Trajectory Tracker

Computer Vision

Blob Detection | Kalman Filter | Parabola Fit | Shot Metrics

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What Does It Do?

🎯 The Application

Given any basketball video, the tracker detects the ball every frame, follows its arc through the air, and draws the full parabolic flight path — all using classical CV algorithms only.

📊 What It Computes

Release angle, arc height, flight time, horizontal displacement, and peak frame index — giving coaches and players objective shot analysis from a simple video clip.

Zero deep learning — every decision is explainable classical CV

System Pipeline

Raw Video → Gaussian Blur → HSV Filter → Blob Detect → Kalman Filter → Shot Segmenter → Trajectory Fit → Output Video

FALLBACK CHAIN

Blob Detection Fails → Shi-Tomasi Corner Detect → Kalman Prediction → Max 8 Frames Predicted

Core Algorithms

Gaussian Blur	Reduce noise before color segmentation
HSV Color Filtering	Isolate orange ball pixels from background
Contour + Circularity	Find round blob candidates, reject non-circular
MOG2 Background Sub	Remove static gym fixtures as false positives
Kalman Filter	Smooth tracking + predict during occlusion
Polynomial Fit (deg=2)	Fit parabola to tracked arc for visualization

5 Iterations of Improvement

1 Baseline Blob Detection

HSV filter + contour detection. Detected many blobs but no filtering — gym fixtures, player skin, and bleachers all triggered false positives.

❌ Too noisy

2 Adaptive HSV Tuning

Sampled real ball pixels using mouse callback tool. Measured H:20-24, S:140-170, V:196-249. Raised saturation floor to S>120 to reject gold bleachers.

⚠️ Better, still false positives

3 ROI + Circularity Filter

Added ROI masking (top 5%, bottom 20% blackout) to remove ceiling fixtures and floor. Raised circularity threshold to 0.72 — only round blobs pass.

✅ False positives gone

4 Kalman + Corner Fallback

Ball disappears behind hoop/net. Added Shi-Tomasi corner fallback and Kalman prediction for up to 8 missed frames. Trajectory no longer breaks.

✅ Occlusion handled

5 Parabola Fit + Smoothing

Raw detections are jittery. Added Gaussian-weighted smoothing + outlier rejection. Fit x=f(y) parabola for vertical shots. Clean arc visualization.

✅ Smooth trajectory

Real-World Failures & How We Addressed Them

❌ Gold Bleacher False Positives

✅ Resolved

Why: Yellow bleacher seats had similar HSV hue to basketball (H~25) and were detected as ball candidates in every frame.

Fix: Raised saturation minimum to S>120. Ball saturation S~140-170 vs bleachers S~80-100. Gap was sufficient to separate them.

❌ Ceiling Fixture Detection

✅ Resolved

Why: Gym ceiling had an orange-colored scoreboard fixture at the top of frame. Tracker locked onto it and ignored the real ball.

Fix: Added ROI blackout: top 5% and bottom 20% of frame zeroed out before blob detection. Ball trajectory never enters these zones.

❌ Ball Behind Hoop / Net

✅ Partially resolved

Why: During arc peak the ball passes through the hoop net causing partial occlusion. Blob detection returns null for several frames.

Fix: Kalman filter predicts position for up to 8 consecutive missed frames using constant-velocity model. Trajectory stays continuous.

❌ Trajectory Not Reaching Hoop

✅ Resolved

Why: ROI top cutoff was too aggressive (15%), blocking the ball near the basket which sits close to the ceiling.

Fix: Reduced roi_top from 0.15 to 0.05. Wider upper detection zone lets the tracker follow the ball all the way to the hoop.

❌ Right-to-Left Shot Direction

✅ Resolved

Why: Release angle computed as arctan2(-dy, dx) returned -166° for right-to-left shots. Shot validator rejected it as impossible.

Fix: Changed to abs(arctan2(-dy, abs(dx))). Absolute value makes angle reflect steepness only — direction doesn't matter.

❌ Ball Too Small at Far Distance

⚠️ Partially resolved

Why: At 3-point range the ball radius is ~8-10px — near the minimum threshold. Detection becomes inconsistent.

Fix: Lowered min_radius to 8px. Still a limitation — very far shots miss several frames. Kalman prediction covers most gaps.

Results & Shot Metrics

21.3°

Release Angle

optimal ~45°

703 px

Arc Height

tracked in frame

4.37 s

Flight Time

frame-accurate

67

Frames Tracked

out of ~130 total

✅ What Worked

- HSV + circularity filter — clean ball isolation
- Kalman smoothing — stable tracking through occlusion
- Parabola fit x=f(y) — correct arc for vertical shots
- State machine segmentation — clean shot boundaries
- ROI masking — eliminated all ceiling false positives

⚠️ Limitations

- Ball too small at 3-point range (<10px radius)
- Trajectory doesn't always reach the hoop fully
- Camera pan/zoom breaks background subtractor
- Multi-shot videos: only 1 of 4 shots detected
- Parameter tuning is scene-specific — not generalizable

Key Takeaway

Classical CV works — but parameter tuning is everything.

Every threshold (circularity=0.72, S_min=120, roi_top=0.05) was chosen by empirical testing on real gym footage — not guesswork.

🔬 **Explainable**

📐 **Tunable**

🏀 **Real-World Tested**